

Reciprocal Preferences and Expectations in International Agreements

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Abstract

This paper investigates the impact of reciprocal preferences and countries' expectations towards each other in the context of international agreements aimed at providing global public goods. Reciprocal countries reward kindness (positive reciprocity) while retaliating against unkind behavior (negative reciprocity). We introduce a model where countries decide not only to participate but also determine the extent of their effort. We demonstrate that reciprocal preferences can exert both positive and negative influences on the effort put forth by the signatories and non-signatories of a treaty, contingent upon their expectations towards others. When reciprocity concerns are close to zero, participation in a stable treaty either remains unchanged or contracts. Nevertheless, even some small reciprocal preferences (i.e., caring for reciprocal concerns as much as about 3.8% of the material payoffs) can lead to the formation of a stable grand coalition with signatories exerting the efficient efforts. Furthermore, we extend our model to a heterogeneous case where reciprocal and self-interested countries coexist. Additionally, we explore a scenario in which a single country strategically and unilaterally employs its expectations towards others.

Keywords: Global Public Goods; Coalition Formation Game; International Agreements; Reciprocity; Expectations

Keywords: F53; H87; Q54

1 Introduction

There is substantial evidence supporting the existence of reciprocal preferences among individuals the desire to reciprocate kindness with kindness and respond to unkind behavior with retaliation (Andreoni, 1988; Camerer, 2003; Charness and Rabin, 2002; Croson, 2007; Dohmen et al., 2009; Falk et al., 2003, 2008; Falk and Fischbacher, 2006; Palfrey and Prisbrey, 1997). If such preferences extend to governments and countries to some extent, how do they impact international agreements aimed at providing global public goods, such as addressing environmental challenges and eradicating epidemics? Additionally, how do countries' expectations towards others, i.e., defining behaviors as kind or unkind, influence these agreements? Finally, how does

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a country behave when strategically employing its expectations towards others? To address these inquiries, we extend and incorporate a model of reciprocal preferences (Rabin, 1993; Segal and Sobel, 2007) into a coalition formation model (Barrett, 1994).

Despite the expectation that reciprocal concerns may be weaker for governments and countries, prior studies present three reasons why these concerns matter in international agreements (Hadjiyiannis et al., 2012; Nyborg, 2018). First, reciprocal concerns are evident in the speeches and declarations of world leaders and policy experts on global issues, including climate change, though one can argue these declarations are made for strategic reasons (Lange et al., 2010). Second, if reciprocal concerns are prevalent among the population, they may also be widespread among those representing countries in negotiations. Third, if the median voter possesses such concerns, governments aiming for reelection may reflect these preferences. We find that even very small concerns for reciprocity can significantly impact international agreements.

To address the ramifications of reciprocal preferences and countries' expectations towards others on international agreements, we employ a coalition formation game (Barrett, 1994), where reciprocal countries face a public goods dilemma and strive to address it by signing a self-enforcing international agreement. The game comprises three stages: first, countries decide whether to sign a treaty or not; then, signatories act collectively to maximize their joint payoff; finally, non-signatories determine their efforts non-cooperatively. We investigate how the reciprocal concerns and countries' expectations influence the effort levels of signatories and non-signatories and the number of signatories in a stable treaty—both in terms of external stability (no new country willing to join) and internal stability (no country willing to leave the treaty). The theoretical international environmental agreements literature (e.g., d'Aspremont et al. (1983); Barrett (1994); Carraro and Siniscalco (1993); Hoel (1992)) have produced a wide range of predominantly negative results, indicating a paradox of cooperation: stable agreements likely to achieve either only a few signatories with fairly ambitious efforts or a large number of signatories with unambitious efforts.¹ Our analysis illuminates whether reciprocal concerns could address this paradox of cooperation.

Our approach aligns with previous studies on the impact of reciprocity on dilemma games. In his seminal paper, Rabin (1993) demonstrated that sufficiently strong reciprocal concerns transform the prisoner's dilemma game into a coordination game where cooperation becomes a Nash equilibrium. Nyborg (2018) incorporates reciprocal preferences (Rabin, 1993; Segal and Sobel, 2007) into a coalition formation game (Barrett, 2003) in which countries' decisions are binary, i.e., to participate or not. Similar to Rabin (1993), first she shows that full participation in the public good becomes a Nash equilibrium if countries have sufficiently strong reciprocal concerns. She finds that both the zero coalition and the grand coalition are stable; the latter requires countries to have sufficiently strong reciprocal preferences. Buchholz et al. (2018) construct a different model with binary decisions. While neither the empty coalition nor the grand coalition is stable under self-interested preferences, both become stable if the countries

¹For a review of this literature, see Barrett (2005) and Finus (2008).

have sufficiently strong reciprocal preferences. Thus, reciprocal preferences can both increase and decrease participation in a stable coalition.²

Our model follows Nyborg (2018)’s framework as close as possible for the sake of tractability and transparency, but depart by introducing continuous effort choices $[0, 1]$ instead of binary $\{0, 1\}$, and quadratic effort costs instead of linear. The latter is needed for the solution to be interior; otherwise, we would have identical results to Nyborg (2018) despite extending the strategy space. These changes would improve the realism of the model since in most international agreements to supply public goods, the decision is not only about participation but also about the degree of contribution.³ On a different note, Lange and Vogt (2003) explored another other-regarding preference, inequality aversion, in coalition formation games. Their study reveal that whether the decision space is binary or continuous yield completely different results on the stability of the grand coalition.

Furthermore, we extend Rabin (1993) by generalizing the threshold for the perception of the kind or the unkind behavior to take any values in $(0, 1]$ rather than only the mid-point, 0.5, which we call this value the fair effort threshold. This allows us to explore the impact of reciprocal preferences under various expectations. Hadjiyiannis et al. (2012) also studies the impact of reciprocal preferences on international environmental agreements with different degrees of expectations towards others. However, rather than a coalition formation game, they study a compliance game. They find that reciprocal countries having high expectations towards others could have detrimental effects on the negotiations and conclude that the failure of the 2009 Copenhagen Summit on climate change might be due to the extremely high expectations known to exist before the meeting.

Reciprocal preferences with continuous effort space yield unique effects, differing from self-interested preferences (Barrett, 1994) and binary decision cases with reciprocal preferences (Nyborg, 2018). Non-signatories, who would have a dominant strategy identical to non-cooperative Nash effort under self-interested preferences, vary their efforts due to reciprocal concerns. Signatories, who perceive others’ efforts on average as (un)kind, exert an effort higher (lower) than the Nash effort. A further increase (decrease) in signatories efforts yields a positive (negative) impact on non-signatories’ effort levels, which yields additional incentives for signatories to raise their efforts.⁴

Our model introduces two thresholds for efforts, one for signatories and one for non-signatories, dividing the value for fair effort threshold into three intervals. In the smallest interval, where countries have low expectations towards others, reciprocity positively impacts

²One important aspect that these models differ is which strategies are perceived as kind or unkind. In Buchholz et al. (2018), countries’ perceptions are self-centered and, thus, they perceive another country’s behavior as (un)kind if this country contributed more (less) than the country itself. On the other hand, Nyborg (2018) follows Rabin (1993) and defines a behavior (un)kind if it yields a higher (lower) payoff than the mid-point of the highest and lowest payoffs that one can secure to the other country.

³In fact, Nyborg (2018) agreed to this point and left it for future research.

⁴This is a surprising effect given that stages 2 and 3 of the coalition formation game resemble a Stackelberg game with signatories playing the role of leaders.

both signatories and non-signatories' efforts as they perceive others as kind. In the highest interval, where countries have high expectations, reciprocity negatively affects both types' efforts as they perceive others as unkind. The middle interval represents moderate expectations in which the impact of reciprocal concerns on signatory and non-signatory efforts differ with signatories' efforts increase and non-signatories' efforts decrease. The reason for this difference is the additional incentive signatories have to increase efforts since they use it strategically to positively influence others.

Regarding the stability of coalitions, Barrett (1994) finds that under the self-interested preferences, for $N \geq 3$, the stable coalition consists of only 3 signatories with the third country being indifferent between joining and remaining outside. Even very small reciprocal preferences have an impact on the internal and external stability conditions and this impact could disincentive this third country to not to join the coalition if the expectations are either low or high. As a result, the stable coalition size contracts. However, the third country is still willing to enter if countries have some moderate expectations from each other. Consequently, the stable coalition size can remain at 3 or contract to 2 at sufficiently small reciprocal concerns.

While solving the general model for not-so-small reciprocal preferences is challenging, our numerical analysis of the condition for an N^{th} country to join, given the other $N - 1$ countries' participation, reveals parameters where the grand coalition is stable. Interestingly, we find that if countries care reciprocal payoffs even as much as about 3.8% of the monetary payoffs, coupled with not too high expectations, then the grand coalition can be stable and countries exert the efficient effort levels. Thus, the results on the stability of the grand coalition found by Nyborg (2018) are robust even under continuous strategy space.

Finally, our paper presents two extensions. First, acknowledging that reciprocal concerns expressed in the international arena may be strategic, we extend our non-cooperative game by assuming that a single country strategically announces its expectations while others remain truthful. We find that when a country strategically uses its expectations, it exaggerates them, providing an excuse to lower its efforts. However, this results in only a marginal decrease in others' efforts. Second, we allow the coexistence of self-interested and reciprocal countries in both non-cooperative and coalition formation games. Our findings reveal that stable coalitions may be nonexistent, unique, or multiple, with stable coalitions comprising solely self-interested or reciprocal countries, or a mix of both. Notably, for sufficiently small α , the stable coalition size may be either 2 or 3, akin to the symmetric case. However, for sufficiently high α , the stable coalition size can match the number of reciprocal countries.

The subsequent sections of the paper are organized as follows. Section 2 introduces the model, while Section 3 examines two extreme cases—non-cooperation and full-cooperation. Section 4 studies a scenario of partial cooperation, where countries decide whether to join a coalition or not. In both Sections 3 and 4, we draw comparisons between the game played by reciprocal countries and that played by self-interested countries. Section 5 introduces two extensions. Detailed calculations, proofs, and additional materials are included in the appendix, with sup-

porting Mathematica file (and its PDF version) available for further clarity and accessibility.

2 Model

Consider a world of $i = 1, \dots, N$ identical countries, where each country i decides an effort level $q_i \in [0, 1]$.⁵ Each country i benefits from its own and also others' efforts but pays its effort cost privately. Following Barrett (1994) (Section 2.2), we assume country i 's material payoff consists of constant marginal benefits and linear marginal costs of effort: $\Pi_i = bQ - \frac{c}{2}q_i^2$, where $Q = \sum_{i=1}^N q_i$ and $b, c > 0$.

In contrast to a self-interested country, a reciprocal country i is concerned not only about its material payoff but also about its reciprocal payoffs:

$$U_i = \Pi_i + \underline{\alpha} R_i, \quad (1)$$

where $\underline{\alpha} > 0$ is a scaling factor measuring how much a country cares about the reciprocal payoffs in comparison to the material payoff. The bilateral reciprocal payoff consists of two parts: first, a psychological (dis)utility of being treated (un)kindly; second, the pleasure of returning the (un)kind behavior. The country i 's reciprocal payoffs R_i is the average of country i 's bilateral reciprocal concerns towards the other $N - 1$ countries. Formally:

$$R_i = \frac{1}{N-1} \left[\sum_{j \neq i} \tilde{f}_{ji} + \sum_{j \neq i} f_{ij} \tilde{f}_{ji} \right], \quad (2)$$

where $f_{ij} > (<) 0$ refers to i being (un)kind towards j and $\tilde{f}_{ji} > (<) 0$ refers to i 's beliefs about j 's (un)kindness towards i ($j \neq i$).⁶ Notice that country i enjoys others being kind toward it, i.e., the first term in the brackets, and also enjoys being (un)kind towards (un)kind, i.e., the second term in the brackets. We follow Rabin (1993) to define i 's kindness toward j :

$$f_{ij} = \frac{\Pi_j(q_i, \tilde{Q}_{-i}) - \Pi_{ij}^e}{\Pi_{ij}^{max} - \Pi_{ij}^{min}}, \quad (3)$$

where Π_{ij}^e be the equitable or fair payoff that i can secure to j , and $\tilde{Q}_{-i} = \sum_{j \neq i} q_j$ is the country i 's belief on other countries' total effort. Furthermore, Π_{ij}^{max} and Π_{ij}^{min} denote the maximum and minimum payoff of country j that i can secure among points that are Pareto efficient in $\Pi_j(\cdot, \tilde{Q}_{-i})$, respectively. Thus, i 's kindness towards j is defined as the difference between the payoff that j can gain based on the i 's strategy and conceivably fair payoff, normalized by the range j 's payoff. This definition implies that the more payoff i secure for j compared to what it is perceived as fair payoff, the kinder (or less unkind) i is perceived toward j .

Barrett (2003) finds the efficient outcome to be $\frac{Nb}{c}$ and we assume it is equal to the upper bound of the strategy space, 1. Despite restricting the strategy space to $[0, 1]$, this assumption

⁵Normalizing the highest effort to 1 significantly simplifies the model. We will adjust some parameters below so that the efficient outcome is 1 and thus all the effort levels of interest lies in $[0, 1]$.

⁶Reciprocal payoffs defined in (2) follows Nyborg (2018) and extends Rabin (1993)'s 2-player normal form game to an N-player game.

guarantees that all the effort levels of interest will be in this strategy space. Barrett (2003) also finds countries have dominant (and thus Nash) strategy to play b/c ; we assume $b < c$ to ensure that countries do not exert full effort when playing non-cooperatively. Also, exerting efforts lower than b/c would be Pareto inefficient. As a result, we have $\Pi_{ij}^{max} = b(\tilde{Q}_{-i} + 1) - \frac{c}{2}q_j^2$ and $\Pi_{ij}^{min} = b(\tilde{Q}_{-i} + \frac{b}{c}) - \frac{c}{2}q_j^2$.⁷ And, $\Pi_{ij}^{max} - \Pi_{ij}^{min} = b(1 - \frac{b}{c})$.

While both Rabin (1993) and Nyborg (2018) define Π_{ij}^e as the midpoint, i.e., $\underline{\eta} = 0.5$, between Π_{ij}^{max} and Π_{ij}^{min} , we extend it as

$$\Pi_{ij}^e = \underline{\eta}\Pi_{ij}^{max} + (1 - \underline{\eta})\Pi_{ij}^{min} = b(\tilde{Q}_{-i} + \underline{\eta} + (1 - \underline{\eta})\frac{b}{c}) - \frac{c}{2}q_j^2 \quad (4)$$

where $\underline{\eta} \in [0, 1]$ is the fair effort threshold, assumed to be identical for all countries. An increase (decrease) in $\underline{\eta}$ means that countries become more (less) demanding towards other countries in terms of what they perceive as fair. Through this extension, we can analyze how countries' expectations (or how demanding they are) toward other countries influence international negotiations to supply a global public good.

From (4) and $\Pi_{ij}^{max} - \Pi_{ij}^{min} = b(1 - b/c)$, f_{ij} is simplified to:

$$f_{ij} = \frac{c(q_i - \underline{\eta})}{c - b}, \quad (5)$$

where $\eta = \underline{\eta}(1 - b/c) + b/c$.⁸ Since $\eta = b/c$ if $\underline{\eta} = 0$ and $\eta = 1$ if $\underline{\eta} = 1$, we have $\eta \in [b/c, 1]$. Based on 5, if $q_i > (<)\eta \iff f_{ij} > (<)0$ and i is perceived to be (un)kind toward j , and if $q_i = \eta \iff f_{ij} = 0$ and i is perceived to be neutral toward j . Similarly, we have $\tilde{f}_{ji} = \frac{c(\tilde{q}_j - \underline{\eta})}{c - b}$. Thus, country i 's reciprocal payoffs can be written as a function of own and others' strategies:

$$R_i = \frac{c}{(c - b)} \left(\frac{\tilde{Q}_{-i}}{N - 1} - \underline{\eta} \right) (1 + q_i - \eta) \quad (6)$$

Notice that the second parenthesis (1 plus how kind i is towards a country j) is always positive. Thus, if country i believes the others' on average are (un)kind, then R_i becomes positive (negative). Consequently, country i experiences positive (negative) reciprocity. Notably, even though a reciprocal country evaluates how fair each country is through its kindness functions, it only has one effort decision that affects all the others. Therefore, its reciprocal payoff depends on the average efforts of the others rather than the individual efforts of others. Combining all, country i 's utility function is expressed as follows:

$$\begin{aligned} U_i &= U(q_i, \tilde{Q}_{-i}) = \Pi_i + \underline{\alpha}R_i \\ &= b(\tilde{Q}_{-i} + q_i) - \frac{c}{2}q_i^2 + \frac{\underline{\alpha}c}{c - b} \left(\frac{\tilde{Q}_{-i}}{N - 1} - \underline{\eta} \right) (1 + q_i - \eta) \\ &= b(\tilde{Q}_{-i} + q_i) - \frac{c}{2}q_i^2 + \alpha \left(\frac{\tilde{Q}_{-i}}{N - 1} - \eta \right) (1 + q_i - \eta). \end{aligned} \quad (7)$$

⁷In Nyborg (2018)'s model with binary strategies, country i would exert no effort at Π_{ij}^{min} .

⁸By using the assumption $bN = c$, we can simplify the kindness function and what it follows. However, we prefer to keep it as is, since it facilitates to understand the differences between our model and the one under self-interested preferences.

So, given that $\frac{c}{c-b} > 0$, we introduce another scaling factor $\alpha = \frac{\alpha c}{(c-b)}$ to simplify notation.

Finally, as Nyborg (2018), we employ Segal and Sobel (2007) in equilibria concepts, which allows countries to have preferences over strategies rather than preferences over beliefs. So, we remove the “tilde”’s.

3 Non-Cooperative and Full-Cooperative Cases

Before turning to coalition formation, we first study the game under two extreme cases: non-cooperative and full cooperative games. In non-cooperative game, given the others’ efforts Q_{-i} , countries maximize their utility $U(q_i, Q_{-i})$ by solving the first-order condition:

$$\frac{\partial U(q_i, Q_{-i})}{\partial q_i} = b - cq_i + \alpha \left(\frac{Q_{-i}}{N-1} - \eta \right) = 0 \quad (8)$$

Solving for q_i under the assumption of symmetry between countries provides a reciprocal country’s non-cooperative effort level:

$$q_{NC}^R = \frac{b - \alpha\eta}{c - \alpha} \quad (9)$$

It’s worth reminding that the non-cooperative effort level under self-interested preferences (i.e., $\alpha = 0$) is $q_{NC}^S = \frac{b}{c}$.

Moving to full cooperation, all N countries jointly maximize their aggregate utilities, which is the sum of all individual utilities. Assuming each country exerts the same effort level, country i ’s utility function would be $U_i = Nbq_i - \frac{c}{2}q_i^2 + \alpha(q_i - \eta)(q_i + 1 - \eta)$. Solving the optimal effort level from the first-order condition: $\frac{\partial U_i}{\partial q_i} = 0$, yields $q_i = \frac{Nb + \alpha(1-2\eta)}{c-2\alpha}$. We assume both $Nb + \alpha(1-2\eta) > 0$ and $c - 2\alpha > 0$ so that full-cooperative effort level is positive. Under the self-interested preferences, the full-cooperative efforts are: $q_{FC}^S = \frac{Nb}{c}$ and we assume $\frac{Nb}{c} = 1$.⁹ Proposition 1 consolidates the non-cooperative and full-cooperative effort levels and elucidates the impact of reciprocal preferences on them.

Proposition 1

- (1) *The non-cooperative Nash efforts under reciprocal preferences are weakly lower than to those under self-interested preferences, i.e., $q_{NC}^R = \frac{b-\alpha\eta}{c-\alpha} \leq q_{NC}^S = \frac{b}{c}$. Furthermore, $q_{NC}^R \geq 0 \iff \alpha\eta \leq b$.*
- (2) *The full-cooperative efforts under reciprocal preferences $q_{FC}^R = 1$.*

Firstly, remember that $\eta \in [\frac{b}{c}, 1]$. By equation (6), if $\eta = b/c$, country i would be neutral to others playing q_{NC}^S and since reciprocity does not play a role, we have $q_{NC}^R = q_{NC}^S$. On the other hand, if $\eta > b/c$ country i would perceive them as unkind and retaliate by lowering its effort, $q_{NC}^R < q_{NC}^S$. We assume $b \geq \alpha\eta$ for the non-cooperative efforts of the game with reciprocal

⁹This assumption and $q_{NC}^S = b/c \in (0, 1)$ imply that in partial cooperation if a signatory country i exerts $q_i = 1$ and the grand coalition (i.e., every country signing the treaty) is stable, then we know that the efficient outcome being implemented by the agreement.

countries to be non-negative. Secondly, the impact of reciprocity on the full-cooperative efforts would be always positive. Since any country would perceive others on average exerting full effort as kind and they would like to return kindness with kindness by increasing their efforts. However, their efforts cannot exceed 1.¹⁰

Nyborg (2018) finds in her model with binary decisions $\{0, 1\}$ that if countries have sufficiently strong reciprocal preferences, then even every country exerting full effort 1 can become the non-cooperative outcome. Thus, sufficiently strong reciprocal preferences convert the public goods dilemma game into a coordination game. However, as we extend the strategy space to the continuous $[0, 1]$, we find this is no longer the case.

4 Partial Cooperation

In this section, we study the properties of stable treaties through a three-stage game, building on Barrett (1994) while introducing reciprocal preferences. The game unfolds as follows:

Stage 1. Every country i decides simultaneously and independently whether to sign or not to sign the treaty. Let k represent the number of signatories.

Stage 2. Signatories collaboratively decide their strategies to maximize their joint payoff.

Stage 3. Non-signatories determine their strategies independently.

We assume a single coalition and solve the game using backward induction. Barrett (1994), under self-interested preferences, reveals that each self-interested non-signatory plays its dominant strategy, identical to the non-cooperative effort levels b/c . Each self-interested signatory exerts an effort level of bk/c . The self-enforcing coalition consists of 2 countries when $N = 2$ and 3 countries when $N \geq 3$. Our study replicates this game, but with all countries having reciprocal preferences. The results are compared with the benchmark model.¹¹

4.1 Non-signatories and Signatories' Efforts

We commence our analysis with the effort levels of non-signatories in stage 3 and notice that stages 2 and 3 resembles the Stackelberg game. For a given coalition size k and signatories' effort level q_s^R , each non-signatory i maximizes the utility function:

$$U_{ni} \equiv b[kq_s^R + (N - k - 1)q_n^R + q_{ni}^R] - \frac{c}{2}q_i^{R2} + \alpha \left(\frac{kq_s^R + (N - k - 1)q_n^R}{N - 1} - \eta \right) (q_{ni}^R + 1 - \eta) \quad (10)$$

¹⁰This is the only case that our effort space affects an outcome, preventing reciprocal countries to exert inefficiently high efforts.

¹¹It's worth noting that as $\alpha \rightarrow 0$, the solutions under reciprocal preferences converge to those under self-interested preferences. Additionally, in Stages 2 and 3, setting $k = N$ yields the full-cooperative outcome, and $k = 0$ yields the non-cooperative outcomes from the previous section.

Solving for the best-response function of reciprocal non-signatory i :

$$\begin{aligned}\frac{\partial U_{ni}}{\partial q_{ni}} &= b - cq_{ni}^R + \alpha \left(\frac{kq_s^R + (N-k-1)q_n^R}{N-1} - \eta \right) = 0 \\ q_{ni} &= \frac{b}{c} + \frac{\alpha}{c} \left(\frac{kq_s^R + (N-k-1)q_n^R}{N-1} - \eta \right)\end{aligned}\quad (11)$$

Notice that it is equal to the effort of the self-interested non-signatory, b/c , plus how much the others exert effort on average in comparison to the fair effort threshold, η . Also, the slope of the best-response function is positive:

$$\frac{\partial q_{ni}}{\partial q_{sj}} = \frac{\partial q_{ni}}{\partial q_{nl}} = \frac{\alpha}{c(N-1)} > 0. \quad (12)$$

Thus, non-signatories' effort is strategic complement of other non-signatories' and signatories' efforts. Next, assuming symmetry, $q_{ni} = q_n \forall i$, by (11) we find non-signatories' best-response:

$$q_n^R(q_s^R) = \frac{b(N-1) + \alpha(kq_s^R - (N-1)\eta)}{c(N-1) - \alpha(N-k-1)} \quad (13)$$

Moving to stage 2, signatories, aware of how non-signatories respond to their efforts, jointly determine their levels to maximize joint utility:

$$U_s = b[kq_s^R + (N-k)q_n^R] - \frac{c}{2}q_s^{R2} + \alpha \left(\frac{(k-1)q_s^R + (N-k)q_n^R}{N-1} - \eta \right) (q_s^R + 1 - \eta) \quad (14)$$

Leaving the detailed calculations to the Appendix, we find the signatories efforts as follows:

$$q_s^R = \frac{bk + \alpha \left(\frac{k-1}{N-1}(1-\eta) + \frac{(N-k)(b-\alpha\eta)}{c(N-1)-\alpha(N-k-1)} - \eta \right) + \left(b(N-k) + \alpha \frac{N-k}{N-1}(1-\eta) \right) \frac{dq_n^R}{dq_s^R}}{c - 2\alpha \left(\frac{k-1}{N-1} + \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} \right)}, \quad (15)$$

where

$$\frac{dq_n^R}{dq_s^R} = \frac{\alpha k}{c(N-1) - \alpha(N-k-1)} > 0 \quad (\because c > 2\alpha.) \quad (16)$$

Let us discuss the impact of reciprocal preferences on the efforts. First, upon incorporating reciprocal preferences, non-signatories' strategies are no longer dominant, becoming contingent on signatories' efforts, see (12) and (16). More interestingly, signatories act strategically, akin to Stackelberg leaders, positively influence non-signatories' efforts by increasing their efforts in the negotiated agreement.¹² In prevailing models of international (environmental) agreements, signatories typically follow one of two patterns: either they refrain from strategic actions due to non-signatories holding dominant strategies ($\frac{dq_n}{dq_s} = 0$), or they strategically decrease their effort levels to incentivize non-signatories to elevate their own efforts ($\frac{dq_n}{dq_s} < 0$). This choice hinges on the particular functional forms governing the benefits and costs, as discussed by Barrett (2005).

The complex expression for q_s^R can be dissected into three components in the numerator. But first, notice that the denominator is positive since the term in the parenthesis is less than

¹²This observation aligns with empirical evidence from trust games and public goods games with leadership, as highlighted by studies such as Cappelen et al. (2016), Gächter and Renner (2018), Güth et al. (2007), and Potters et al. (2007). In the context of a public goods game with leadership, the leader initiates by making the first move, and subsequent players make decisions after observing the leader's actions.

1 and we assume $c > 2\alpha$. The first component in the numerator represents material payoffs; recall that bk/c is the signatories' effort level under self-interested preferences. The second component captures the first-order effect of a signatory's reciprocal preferences towards the other $k - 1$ signatories and $N - k$ non-signatories, which can be either positive or negative contingent on the fair effort threshold η . Notably, η always enters negatively in the second term, meaning that as η increases, i.e., some efforts that signatories used to perceive as kind are perceived as unkind now, signatories will lower their efforts. Finally, the third term reflects the second-order impact on $N - k$ non-signatories, always positive and reinforcing the reciprocal cycle.

Employing Mathematica, we could integrate (16) and simplify (15) as follows:

$$q_s^R = \frac{bck(N-1) + \alpha(bN + \alpha(1-2\eta) + c(k-1-\eta(N+k-2)))}{c^2(N-1) - \alpha c(N+k-3) - 2\alpha^2} \quad (17)$$

Plugging (17) into (13) yields non-signatories' effort:

$$q_n^R = \frac{b(N-1) + \alpha(k \left(\frac{bck(N-1) + \alpha(bN + \alpha(1-2\eta) + c(k-1-\eta(N+k-2)))}{c^2(N-1) - \alpha c(N+k-3) - 2\alpha^2} \right) - (N-1)\eta)}{c(N-1) - \alpha(N-k-1)} \quad (18)$$

Proposition 2 (On Effort Levels)

- (1) Both effort levels q_n^R and q_s^R are monotonically increasing in the coalition size k .
- (2) The order of effort levels: $q_{NC}^R \leq q_n^R \leq q_s^R$.

First, notice that in the game with self-interested preferences, non-signatories effort levels do not change in k ($q_n^S = b/c$), but signatories' effort level increases ($q_s^S = bk/c$). Upon introducing reciprocal concerns, an increase in treaty participation boosts the average effort levels of both signatories and non-signatories and thus fosters positive reciprocal responses.

The second point compares the effort levels of signatories and non-signatories in the coalition formation game with the non-cooperative Nash effort level. Unsurprisingly, we have $q_s^R \geq q_n^R$ by the structure of the game. Moreover, $q_n^R \geq q_{NC}^R$ because in the latter all countries exert the same non-cooperative effort level, but in the former there are some signatories exerting a higher effort level positively affecting non-signatories reciprocal payoffs, which results in kinder response.

Now, let us investigate the impact of the reciprocal preferences $\underline{\alpha}$ on the effort levels of both signatory and the non-signatory countries. To explore this, we'll analyze the partial derivatives of effort levels concerning $\underline{\alpha}$ and assess them at $\underline{\alpha} \approx 0$, operating under the assumption that reciprocal concerns are relatively small.¹³ Proposition 3 shows how reciprocal concerns influence effort levels based on the countries' expectations from one another, η .

¹³Regrettably, a comprehensive analysis of the impact of $\underline{\alpha}$ for any arbitrary value proves to be intricate. However, this assumption aligns with Rabin (1993), characterizing reciprocal concerns to be smaller as the stakes become larger. Given the substantial material payoffs in international agreements, this assumption seems reasonable. Still, we'll explore how higher reciprocal concerns might alter our results towards the end of Section 4.

Proposition 3 (*Impact of reciprocity on effort levels*):

$$(1) \left. \frac{\partial q_s^R}{\partial \alpha} \right|_{\alpha=0} \begin{cases} > 0 & \text{if } 0 < \eta < T_H \\ \leq 0 & \text{if } T_H \leq \eta \leq 1 \end{cases}, \text{ where } T_H \equiv \frac{b(N(k+1)+k(k-3))+c(k-1)}{N+k-2},$$

$$(2) \left. \frac{\partial q_n^R}{\partial \alpha} \right|_{\alpha=0} \begin{cases} > 0 & \text{if } 0 < \eta < T_L \\ \leq 0 & \text{if } T_L \leq \eta \leq 1 \end{cases}, \text{ where } T_L \equiv \frac{b(N-k-1+k^2)}{c(N-1)} < T_H.$$

Furthermore, $T_H > 1$ if k is sufficiently high and $T_L > 1$ if $k = N$.

Proposition 3 introduces two pivotal thresholds for η when α is sufficiently small, T_H for signatories and T_L for non-signatories. The positioning of η relative to these thresholds determines whether countries hold high ($\eta > T_H$), moderate ($T_L \leq \eta \leq T_H$), or low ($\eta < T_L$) expectations regarding the efforts of other nations in providing the public good. When expectations are low (high), the impact of reciprocity α on both signatory and non-signatory effort levels is positive (negative). The main driver of these results is as they perceive others on average (un)kind and increase (decrease) efforts to retaliate. If expectations fall within the moderate range, the effect on signatory's effort levels is positive, while on non-signatory's effort levels is negative. The main driver of this difference comes from the fact we found earlier: on the contrary to non-signatories, signatories act strategically and positively influence non-signatories' efforts by increasing (or decrease less) their efforts. Finally, notice that both thresholds depend on k and in fact increase with k . We find that the thresholds can reach to 1. For instance, for $N = 100$, if $k \geq 62$, then $T_H > 1$. In such a case, the high expectation interval would disappear. On the other hand, $T_L > 1$ could only hold if every country participates in the coalition $k = N$ and the impact of reciprocity on efforts is positive for all.

4.2 Stable Treaty

In this subsection, we study the impact of reciprocal preferences on the self-enforcing stable treaty participation; its size is denoted by k^* (Stage 1). To this end, we find the indirect utility functions of signatories and non-signatories as U_s and U_{ns} by substituting q_s^R and q_n^R in their respective utility functions (14) and (10).¹⁴ A coalition achieves stability when no country wishes to either join or leave the coalition. Formally:

Definition 1 *A coalition of size k is stable if $U_s(k) \geq U_{ns}(k-1)$ (i.e., internal stability) and $U_s(k+1) \leq U_{ns}(k)$ (i.e., external stability).*

While the indirect utility functions monotonically increase under self-interested preferences for $k > 1$, Lemma 1 states the properties of the indirect utility functions under reciprocal preferences.¹⁵

¹⁴We present the indirect utility functions only in our supporting Mathematica file due to their size.

¹⁵Although our model substantially diverges from Diamantoudi and Sartzetakis (2006), their Proposition 2 inspired us in Lemma 1.

Lemma 1 Let $U_{ns}(k)$ and $U_s(k)$ respectively be the indirect utility functions of non-signatories and signatories and let $z^{min} = 1 - \frac{\alpha}{c}$. Then,

- (1) $z^{min} := \arg \min_{k \in [0, N]} U_s(k)$.
- (2) U_s increases in k if $k > z^{min}$ and decreases if $k < z^{min}$.
- (3) $U_{ns}(k) < U_s(k)$ for all $0 \leq k < z^{min}$, and $U_{ns}(k) > U_s(k)$ for all $k > z^{min}$.
- (4) $U_{ns}(z^{min}) = U_s(z^{min})$.
- (5) $U_{ns}(k)$ increases for $k > z^{min}$.

Compared to the model under self-interested preferences, the participation level that gives identical indirect utility level to both signatories and non-signatories, i.e., z^{min} , decreases due to reciprocal preferences. For $k > z^{min}$ both $U_{ns}(k)$ and $U_s(k)$ increase with $U_{ns}(k) > U_s(k)$.

Next, we study how the reciprocal preferences affect the indirect utility functions. In all that follows, we continue to assume $\underline{\alpha}$ is sufficiently small.

Lemma 2 (*Impact of Reciprocity on Indirect Utility Functions*) For sufficiently small $\underline{\alpha}$:

- (1) $\left. \frac{\partial U_s(k)}{\partial \underline{\alpha}} \right|_{\underline{\alpha}=0} \begin{cases} > 0 & \text{if } 0 \leq \eta < T_L \\ \leq 0 & \text{if } T_L \leq \eta \leq 1 \end{cases}$
- (2) $\left. \frac{\partial U_{ns}(k)}{\partial \underline{\alpha}} \right|_{\underline{\alpha}=0} \begin{cases} > 0 & \text{if } 0 \leq \eta < T_{U_{ns}} \\ \leq 0 & \text{if } T_{U_{ns}} \leq \eta \leq 1 \end{cases}$
- (3) $\left. \frac{\partial U_{ns}(k-1)}{\partial \underline{\alpha}} \right|_{\underline{\alpha}=0} \begin{cases} > 0 & \text{if } 0 \leq \eta < T_{U_{ns}(k-1)} \\ \leq 0 & \text{if } T_{U_{ns}(k-1)} \leq \eta \leq 1 \end{cases}$.
- (4) *Thresholds:*
 - $T_{U_{ns}(k-1)} < T_L < T_{U_{ns}}$ if $1 \leq k \leq 2$
 - $T_L < T_{U_{ns}(k-1)} < T_{U_{ns}}$ if $k > 2$

Lemma 2 reveals that the impact of reciprocal preferences $\underline{\alpha}$ on the indirect utility functions at the participation level k mirrors the one on the effort level. Notice from point 4) that the order of the thresholds differs whether $k \leq 2$ or $k > 2$. Focusing on the case with $k > 2$: there are three critical thresholds for η , T_L , $T_{U_{ns}(k-1)}$, and $T_{U_{ns}}$ help interpret countries' expectations towards others as high ($\eta > T_{U_{ns}}$), high-moderate ($T_{U_{ns}} \geq \eta \geq T_{U_{ns}(k-1)}$), low-moderate ($T_{U_{ns}(k-1)} > \eta \geq T_L$), and low as before ($\eta > T_L$). Notice that the space defined for expectations slightly differs from the ones we used for effort levels in Proposition 3: see Figure 1, which shows $\underline{\alpha}$'s impact on the effort levels on top, and on indirect utility functions at the bottom part under the case $k \approx 3$ and $N \geq 5$. High (low) expectations yield a positive (negative) impact of reciprocity on all indirect utility functions. As expectations increase, the impact of α on $U_s(k)$, $U_{ns}(k)$, and $U_{ns}(k-1)$ becomes negative. Notice that in Lemma 2, we held k constant and thus it is both

signatories and non-signatories' effort levels that drive this result. With reciprocal concerns α , as expectations increase, both types' efforts tend to decrease, which results in a decrease in indirect utility functions.

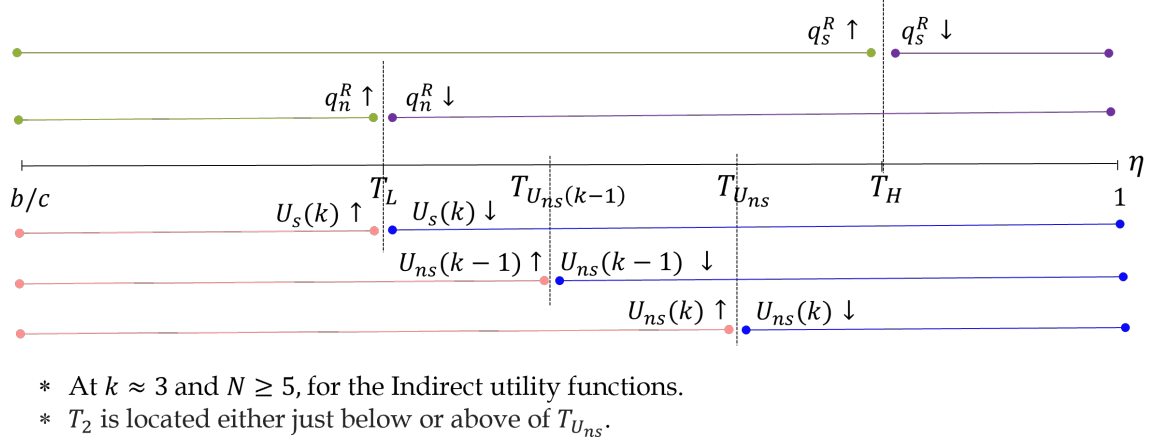


Figure 1: The Impact of Small $\underline{\alpha}$ on efforts and indirect utility functions

Proposition 4 reveals observations about the stable treaty under reciprocal preferences.

Proposition 4 *If $\underline{\alpha}$ is sufficiently small, then the stable treaty size*

1. *Contracts to 2 if countries have either low to low-moderate expectations, $\eta < T_{U_{ns}(k-1)}$ or relatively high expectations, $\eta > T_2 \equiv \frac{2N+4}{N^2}$;*
2. *Remains at 3 if they have relatively moderate expectations, $T_{U_{ns}(k-1)} < \eta < T_2$.*

Furthermore, around $k \approx 3$, i) $T_2 > T_{U_{ns}(k-1)}$, and ii) $T_2 < T_{U_{ns}}$ for $N \leq 17$ and $T_2 > T_{U_{ns}}$ otherwise.

Under self-interested preferences, the stable coalition size is always 3 with the third country being indifferent between joining or not, i.e., $U_n^S(k-1) = U_s^S(k)$ at $k = 3$. However, with the introduction of reciprocal preferences, this size can decrease but does not increase for sufficiently small reciprocal concerns. The determining factor is the internal stability condition, reliant on the intersection of $U_{ns}(k-1)$ and $U_s(k)$. With the introduction of $\underline{\alpha}$, the indirect utility curves could shift in the direction that is indicated in Lemma 2 and Figure 1. For instance, focusing on $k > 2$, for $\eta \in (T_L, T_{U_{ns}(k-1)})$, the impact of $\underline{\alpha}$ on U_s is negative and positive on $U_{ns}(k-1)$, which suffices the stable coalition size to contract to 2. For the other η values, these two indirect utility functions move in the same direction. So, in order to understand how $\underline{\alpha}$ affects the stable coalition size for other η 's, we analyze which of the indirect utility functions, $U_s(k)$ and $U_{ns}(k-1)$, changes faster with $\underline{\alpha}$ around $k \approx 3$. We find that the impact of α on $U_{ns}(k-1)$ is larger than the one on $U_s(k)$ (i.e., $\frac{\partial U_{ns}(k-1)}{\partial \underline{\alpha}} > \frac{\partial U_s(k)}{\partial \underline{\alpha}}$) at $\eta < T_2$, and smaller otherwise. Furthermore, while $T_2 > T_{U_{ns}(k-1)}$ always holds, whether T_2 is higher or lower than $T_{U_{ns}}$ depends on the number of countries. These imply that at $\eta < T_L$, where all indirect utility functions increase in $\underline{\alpha}$, since $U_{ns}(k-1)$ increases faster than $U_s(k)$ around $k \approx 3$, the stable

coalition would contract to 2. At $\eta > T_2$, where both $U_s(k)$ and $U_{ns}(k-1)$ decrease in $\underline{\alpha}$, since $U_s(k)$ decreases faster than $U_{ns}(k-1)$ with α around $k \approx 3$, the stable coalition would again contract to 2. Finally, at $T_{U_{ns}(k-1)} < \eta < T_2$, where both $U_s(k)$ and $U_{ns}(k-1)$ decrease in $\underline{\alpha}$, since $U_{ns}(k-1)$ decreases faster than $U_s(k)$ around $k \approx 3$, the stable coalition size remain at 3.

While Proposition 4 gives rather pessimistic news, we must remember that the reason behind the inability to achieve a positive impact on cooperation may stem from the assumption of sufficiently small α . To delve deeper, we explore whether any feasible parameter values exist that yield a stable grand coalition. To this end, we examine the condition for the N^{th} country to sign the treaty if the other $N-1$ countries have already done so, expressed as $U_s(N) \geq U_{ns}(N-1)$. This simpler inequality, free from the sufficiently small $\underline{\alpha}$ assumption, will allow us to explore higher $\underline{\alpha}$ values through numerical analysis. The aim is to determine whether stable grand coalitions are achievable under this condition and, if achievable, what is the smallest $\underline{\alpha}$ value that allows the stability of the grand coalition.¹⁶

Result 1 (*Numerical Analysis*) *For some parameter values (α, c, η, N) , countries exert the efficient full efforts and the grand coalition is stable. The parameter values with the smallest α that still yield a stable grand coalition is $(\alpha = 0.038, c = 0.1, \eta < 0.8, N \in [4, 100])$.*

As the result indicates, for some parameter values the grand coalition is stable and we find that the smallest value of $\alpha = 0.038$ at which the grand coalition is stable for any $N \geq 4$, $\eta \leq 0.8$, and $c = 0.1$. So, the sufficiently large α needed for the stability of grand coalition is in fact reasonably low. To put differently, for $c = 0.1$, countries caring for reciprocal preference only about 3.8% (more precisely, $\alpha = 0.038 \implies \underline{\alpha} = \frac{3.8\%N}{N-1}$) of their material payoffs with not very high expectations suffice for the stability of the grand coalition.¹⁷

By increasing both $\underline{\alpha}$ and c , we continue to find the other conditions at which the grand coalition is stable.¹⁸ As we increase $\eta \geq 0.8$, we observe condition $U_s(N) \geq U_{ns}(N-1)$ fails to hold, starting with the highest N values: e.g., if $N = 100$, then the condition does not hold anymore as we increase $\eta \geq 0.8$ but it continues to hold if the number of countries is $N = 50$. In all cases that we could find that the condition $U_s(N) \geq U_{ns}(N-1)$ holds, the N^{th} country is indifferent between joining or not.

Finally, given that countries could attain a stable grand coalition and exert the efficient full efforts for some parameters, sufficiently high reciprocal concerns could address the paradox of cooperation.

¹⁶We thank Andreas Lange for suggesting this analysis.

¹⁷We defined $\alpha = \frac{\alpha c}{c-b}$. Substituting b using the assumption $bN/c = 1$ yields $\alpha = \frac{\alpha N}{N-1}$.

¹⁸While there are abundance of other parameter values, here are two other: $(\alpha = 0.438, c = 1, \eta < 0.8, N \in [6, 100])$, $(\alpha = 0.8, c = 2, \eta < 0.8, N \in [5, 100])$

5 Extensions

5.1 Strategic Use of Expectations

In the preceding sections, we operated under the assumption that the fair effort threshold η is exogenous and identical to all countries. However, a pertinent question emerges: what factors would determine η for an individual country? While addressing this empirical question falls beyond the scope of our paper, we investigate how a country, fueled by reciprocal preferences, might strategically wield its expectations toward others.

Instead of navigating the intricacies of a coalition formation game, we simplify the scenario by considering a non-cooperative game.¹⁹ Here, countries disclose their fair effort thresholds in the initial stage and determine their effort levels in the subsequent stage. To comprehend the individual incentive behind choosing a fair effort threshold, we introduce an assumption: $N - 1$ countries truthfully declare their fair effort threshold η_T , while a single country strategically reveals its η_S to maximize its utility, while its true fair effort threshold is also η_T . For simplicity, we assume that countries believe others' fair threshold expectations to be true.²⁰ Due to the model's complexity, we turn to numerical analysis to gain insights. We employ parameter values $c \in (0, 10]$, $\alpha \in (0, 1]$, and $N \in \{5, \dots, 100\}$, while adhering to the previously stated assumptions, $b < c$, $c > 2\alpha$, and $bN/c = 1$. Two distinct assumptions guide our analysis:

Assumption 1

1. *Self-fulfilling prophecy: The strategic country S also believes its announcement η_S to be true.*
2. *Self-awareness: The strategic country S is aware of its true parameter value η_T , but strategically declares η'_S .*

Under the self-fulfilling prophecy assumption, country S maximizes the following utility function:

$$U_S = b((N - 1)q_T + q_S) - \frac{c}{2}q_S^2 + \alpha(q_T - \eta_S)(q_S + 1 - \eta_T). \quad (19)$$

We similarly define the utility function of the truthful countries U_T . Solving the game backward, we determine the best-response functions for q_S using U_S and q_T using U_T . Substituting one into the other yields the effort levels:

$$q_S(\eta_T, \eta_S) = \frac{b(c(N - 1) + \alpha) - \alpha^2(N - 1)\eta_T - ((N - 1)(c - \alpha) + \alpha^2)\eta_S}{(c - \alpha)(c(N - 1) + \alpha)} \quad (20)$$

$$q_T(\eta_T, \eta_S) = \frac{b(c(N - 1) + \alpha) - \alpha(c(N - 1)\eta_T + \alpha\eta_S)}{(c - \alpha)(c(N - 1) + \alpha)} \quad (21)$$

¹⁹Unfortunately, the coalition formation game becomes too complex to analyze even numerically.

²⁰While acknowledging the strength and potential unrealistic nature of this assumption—it overlooks concerns about the credibility of announcements—our simplified model serves to elucidate the directional changes and determinants of the strategically chosen η_S . Introducing concerns for the credibility of the announcement, in the form of a cost for deviating from the true η_T , would quantitatively impact the results without altering their qualitative nature.

Two noteworthy points emerge from the effort levels. Firstly, both fair threshold concerns, η_T and η_S , negatively impact the effort levels. Secondly, each type's own fair effort threshold decreases its own effort level faster than the other type's fair effort threshold.

Apart from the effort levels, η_S appears in the utility function (19). Regardless of the effort levels, an increase in η_S also decreases utility, as country S perceives others as more unkind or less kind. These three forces determine the strategically chosen η_S . On the one hand, by strategically setting $\eta_S > \eta_T$, country S could find an excuse to lower its effort, significantly decreasing the individual cost, while a higher η_S only marginally lowers other countries' efforts. On the other hand, there is an incentive to lower η_S since it helps perceive others as kinder.

In Stage 1, we find the indirect utility function of country S , $U_S(q_S(\cdot), q_T(\cdot))$ by substituting these effort levels into the utility function (19). Solving for η_S from the first order condition $\frac{U_S(q_S(\cdot), q_T(\cdot))}{\eta_S} = 0$ yields:

$$\eta_S = X(b, c, \alpha, N) - Y(b, c, \alpha, N)\eta_T, \quad (22)$$

where both $X(\cdot)$ and $Y(\cdot)$ are positive for all parameter values and, thus, the true expectation η_T always (weakly) decreases the strategically chosen expectation η_S . When we don't enforce the assumption $bN = c \iff b = c/N$, the number of countries N does not affect $X(\cdot)$ or $Y(\cdot)$. When it is enforced and hence also determines b for any given c , it reduces the intercept term $X(\cdot)$. The reciprocal concern α decreases both $X(\cdot)$ and $Y(\cdot)$. Though c affects both $X(\cdot)$ and $Y(\cdot)$, its main impact is to define which α 's satisfy the assumption $c > 2\alpha$. Furthermore, by Proposition 1, we know that $\alpha\eta \leq b \iff \alpha\eta N \leq c$ needed for the Nash efforts to be non-negative. Similarly, we need significant discrepancy between α and c to make sure that the efforts of the truthful and strategic countries are non-negative.

Result 2 (*Self-fulfilling prophecy*) *For all parameter values that satisfy $q_S, q_T \in (0, 1]$, country S sets $\eta_S = 1$ and perceives others as unkind.*

Result 2 summarizes the findings under the self-fulfilling prophecy assumption. Country S always sets its expectations higher than the others $\eta_S = 1 > \eta_T$, unless the true expectations are also 1. Thus, the effect of finding the excuse to lower own efforts (and costs) would be the dominant effect.

Now, let's study how our analysis transforms under the self-awareness assumption. In this scenario, other countries believe the fair effort threshold announced by country S , η'_S . Consequently, their effort level becomes $q_T(\eta_T, \eta'_S)$, mirroring the structure in Equation (21). Anticipating this, country S optimally sets $q_S(\eta_T, q_T(\eta_T, \eta'_S))$, aligning with $q_T(\eta_T, \eta'_S)$ while maintaining its true fair effort threshold at η_T . Both η_T and η'_S negatively impact the effort levels.

In contrast to the analysis under the self-fulfilling prophecy assumption, the strategically chosen η'_S has no direct impact on the utility. Country S is aware that its true fair effort threshold is η_T and evaluates utility accordingly. Absent any force decreasing the strategically chosen η'_S , country S consistently leverages η'_S to find an excuse to lower its effort level by setting $\eta'_S = 1 \geq \eta_T$.

Result 3 (*Self-awareness*) For all parameter values that satisfy $q_S, q_T \in (0, 1]$, country S sets $\eta'_S = 1$ and perceives others as unkind.

How would these results differ in our coalition formation game? The brief answer is simply we do not know. However, let us point out that the objective of countries in non-cooperative game and non-signatories are quite parallel with only difference is signatories exert a higher effort level, increasing overall effort level. Thus, we expect strategic use of expectation to be similar for the non-signatories. However, it is harder to predict how it would affect signatories as their objective is to maximize their joint utilities.²¹

5.2 Heterogeneity: Self-interested and Reciprocal Countries

Let us dive into the realm of heterogeneity, where countries harbor diverse preferences—some purely self-interested and others have reciprocal concerns. The flavor of the game changes with preferences now denoted as:

$$U_i = \Pi_i + \alpha_i R_i, \quad (23)$$

where $\alpha_i \in \{0, \alpha\}$. Here, $A < N$ represents the number of countries with $\alpha_i = \alpha$ and $N - A$ denotes those with $\alpha_i = 0$.

Proposition 5 (*Effort Levels in Non-Cooperative Game*)

(1) *Self-interested and Reciprocal Countries' Nash Efforts:*

$$q_{NC}^{HS} = \frac{b}{c}; \quad q_{NC}^{HR} = \frac{bc(N-1) + b(N-A)\alpha - c(N-1)\alpha\eta}{c(c(N-1) + \alpha - A\alpha)}$$

(2) *The impact of reciprocal concerns on Nash Efforts:*

$$\frac{\partial q_{NC}^{HR}}{\partial \alpha} \leq 0 \iff \eta \geq \frac{b}{c} = q_{NC}^{HS}$$

For sufficiently high α , reciprocal countries could exert zero if A and η are large enough.

Proposition 5 unfolds the dynamics of Nash efforts—when reciprocal countries anticipate higher efforts than self-interested counterparts, they exert less. And if the reciprocal concerns are strong enough, then reciprocal countries could exert no effort at all.

Subsequently, we engage in the resolution of the coalition formation game featuring heterogeneous countries, supplemented by a numerical analysis to extract meaningful insights. The intricate calculations, akin to those in the symmetric scenario, are available in the accompanying Mathematica file. Within this heterogeneous context, there are four stability conditions: two for self-interested and two for reciprocal countries. We summarize our findings below:

Result 4 (*Numerical Analysis*)

²¹In an ongoing project, İriş studies the strategic use of expectations in coalition formation game when the decisions are binary.

1. *Number: Stable coalitions may be non-existent, unique, or many.*
2. *Formation: The stable coalitions may comprise exclusively self-interested countries, entirely reciprocal countries, or a mix of both. The reciprocal concerns α has to be sufficiently small for both types of countries to comprise a stable coalition jointly.*
3. *Expectations: As η increases, the stable coalition tilts towards more self-interested and fewer reciprocal countries.*
4. *Size: A small α yields stable coalition of size $k^* \in \{2, 3\}$. A sufficiently high α coupled with sufficiently low η , up to A reciprocal countries can form a stable coalition.*

6 Conclusion

In this article, we have examined the impact of reciprocal preferences on countries' efforts and participation in the provision of public goods. Through an extension of a coalition formation model that incorporates reciprocal preferences, we unveil the pivotal role played by countries' expectations towards each other in shaping the outcomes of positive and negative reciprocity. Specifically, when expectations are low, both signatories and non-signatories increase their efforts in response to reciprocal concerns. Conversely, elevated expectations prompt a reduction in efforts from both groups. In situations of moderate expectations, signatories increase their efforts, while non-signatories reduce theirs. This difference arise due to signatories acting strategically and positively influence non-signatories by increasing their own efforts.

The implications for climate change negotiations are noteworthy. The positive influence of signatories' increased efforts on non-signatories could encourage more cooperation. An example of signatories' higher efforts can be EU's leading efforts in curbing emissions that might be inspiring other countries to heighten their efforts. An example of non-signatories' positive response to signatories' increasing efforts can be the US increasing its efforts to tackle climate change despite leaving the Kyoto Protocol.

Moreover, while sufficiently small concerns for reciprocity yield the stable coalition size to either contract or remain the same depending on the countries' degree of expectations, sufficiently strong concerns result in a stable grand coalition with countries exerting the efficient effort levels. We numerically found that this sufficiently strong concerns can be as low as about 3.8% of the material payoffs and, thus, arguably reasonable to consider. Therefore, extending the strategy space so that countries not only decide to participate or not but also decide how much effort to exert produces similar results to the models that focus on participation decision only (Nyborg, 2018; Buchholz et al., 2018). Furthermore, it also addresses the paradox of cooperation.

We have also extended our model to consider the strategic use of expectations and the coexistence of self-interested and reciprocal countries. First, we find when a country strategically uses its expectations in non-cooperative game, it exaggerates its expectations towards others,

allowing it to find an excuse to lower its efforts and, thus, its cost. But this only slightly reduces others' efforts. Second, the presence of both types of countries may result in many stable coalitions of varying sizes, which is up to the number of reciprocal countries.

Overall, our study highlights the importance of reciprocal concerns and expectations in determining countries' participation and efforts in the provision of global public goods. By incorporating these factors into coalition formation models, policymakers and negotiators may gain greater insights into the dynamics of international cooperation and the factors that promote or hinder it.

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Declaration of Generative AI: During the preparation of this work, we used ChatGPT to improve the readability of the paper. After using this tool, we reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix: Proofs and Calculations

A.1 Calculations

A.1.1 Reciprocal Preferences

- Obtaining i 's kindness towards j , the equation (5):

$$\begin{aligned} f_{ij} &= \frac{b(\tilde{Q}_{-i} + q_i) - \frac{c}{2}q_i^2 - (b(\tilde{Q}_{-i} + \underline{\eta} + (1 - \underline{\eta})\frac{b}{c}) - \frac{c}{2}q_i^2)}{b(1 - \frac{b}{c})} \\ &= \frac{q_i - \underline{\eta} - (1 - \underline{\eta})\frac{b}{c}}{(1 - \frac{b}{c})} = \frac{q_i - \underline{\eta}(1 - b/c) - b/c}{(1 - \frac{b}{c})} \end{aligned} \quad (\text{A1})$$

Using $\eta = \underline{\eta}(1 - b/c) + b/c$, we can rewrite f_{ij} :

$$f_{ij} = \frac{q_i - \eta}{(1 - \frac{b}{c})} = \frac{c(q_i - \eta)}{c - b} \quad (\text{A2})$$

- Obtaining i 's reciprocal preferences, the equation (6):

$$\begin{aligned} R_i &= \frac{1}{N-1} \left[\sum_{j \neq i} \left(\frac{c(\tilde{q}_j - \eta)}{c - b} \right) + \sum_{j \neq i} \left(\frac{c(q_i - \eta)}{c - b} \right) \left(\frac{c(\tilde{q}_j - \eta)}{c - b} \right) \right] \\ &= \frac{c}{(c - b)(N-1)} \left[\left(\tilde{Q}_{-i} - (N-1)\eta \right) (1 + q_i - \eta) \right] \\ &= \frac{c}{(c - b)} \left(\frac{\tilde{Q}_{-i}}{N-1} - \eta \right) (1 + q_i - \eta) \end{aligned} \quad (\text{A3})$$

A.1.2 Signatories' Efforts

Using (14) and signatories taking non-signatories' response into account, we find the first order condition:

$$\begin{aligned} \frac{\partial U_s}{\partial q_s^R} &= bk + b(N - k) \frac{dq_n^R}{dq_s^R} - cq_s^R \\ &\quad + \alpha \left[\left(\frac{(k-1)q_s^R + (N-k)q_n^R}{N-1} - \eta \right) + (q_s^R + 1 - \eta) \left(\frac{k-1}{N-1} + \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} \right) \right] \\ &= \left[\alpha \left(2 \frac{k-1}{N-1} + \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} \right) - c \right] q_s^R \\ &\quad + bk + b(N - k) \frac{dq_n^R}{dq_s^R} + \alpha \left[\frac{N-k}{N-1} q_n^R - \eta + (1 - \eta) \left(\frac{k-1}{N-1} + \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} \right) \right] = 0 \end{aligned} \quad (\text{A4})$$

Note that we can rewrite non-signatories' best-response as follows:

$$q_n^R = \frac{dq_n^R}{dq_s^R} q_s^R + \frac{(N-1)(b - \alpha\eta)}{c(N-1) - \alpha(N-k-1)}$$

Multiplying both sides with $\alpha \frac{N-k}{N-1}$:

$$\begin{aligned} \alpha \frac{N-k}{N-1} q_n^R &= \alpha \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} q_s^R + \alpha \frac{N-k}{N-1} \cdot \frac{(N-1)(b - \alpha\eta)}{c(N-1) - \alpha(N-k-1)} \\ &= \alpha \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} q_s^R + \alpha \frac{(N-k)(b - \alpha\eta)}{c(N-1) - \alpha(N-k-1)} \end{aligned}$$

Integrating this term to the equation (A4), we have

$$\begin{aligned}
\frac{\partial U_s}{\partial q_s^R} &= \left[\alpha \left(2 \frac{k-1}{N-1} + \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} \right) - c \right] q_s^R \\
&\quad + bk + b(N-k) \frac{dq_n^R}{dq_s^R} + \left(\alpha \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} q_s^R + \alpha \frac{(N-k)(b-\alpha\eta)}{c(N-1) - \alpha(N-k-1)} \right) \\
&\quad + \alpha \left(-\eta + \frac{k-1}{N-1} (1-\eta) \right) + \alpha \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} (1-\eta) \\
&= \left[2\alpha \left(\frac{k-1}{N-1} + \frac{N-k}{N-1} \cdot \frac{dq_n^R}{dq_s^R} \right) - c \right] q_s^R + \left(b(N-k) + \alpha \frac{N-k}{N-1} (1-\eta) \right) \frac{dq_n^R}{dq_s^R} \\
&\quad + bk + \alpha \left[\frac{k-1}{N-1} (1-\eta) + \frac{(N-k)(b-\alpha\eta)}{c(N-1) - \alpha(N-k-1)} - \eta \right] = 0
\end{aligned}$$

Finally, leaving q_s^R alone and reorganizing the terms:

$$q_s^R = \frac{bk + \alpha \left[\frac{k-1}{N-1} (1-\eta) + \frac{(N-k)(b-\alpha\eta)}{c(N-1) - \alpha(N-k-1)} - \eta \right] + \left(b(N-k) + \alpha \frac{N-k}{N-1} (1-\eta) \right) \frac{dq_n^R}{dq_s^R}}{c - 2\alpha \left(\frac{k-1}{N-1} + \frac{N-k}{N-1} \frac{dq_n^R}{dq_s^R} \right)}$$

A.1.3 Stable treaty under Standard game

Under the self-interested game where $\alpha = 0$, the self-interested indirect utility functions for signatories and non-signatories are respectively:

$$U_{ns}^S(k) = \frac{b^2(2k(k-1) + 2N-1)}{2c}; \quad U_s^S(k) = \frac{b^2(k(k-2) + 2N)}{2c} \quad (\text{A5})$$

The two intersections of $U_s^S(k)$ and $U_{ns}^S(k-1)$ are at $k=1$ and $k=3$. ($U_s^S(k) = U_{ns}^S(k-1) \Leftrightarrow \frac{b^2(k(k-2)+2N)}{2c} = \frac{b^2(2(k-1)(k-2)+2N-1)}{2c} \Leftrightarrow \frac{b^2}{2c}(k-1)(k-3) = 0$). Then the stable coalition size is 3 in this case.

A.2 Proofs

Proof of Proposition 1.

- (1) The non-cooperative efforts of the countries with reciprocal preferences are higher than the ones with self-interested preferences if

$$\frac{b-\alpha\eta}{c-\alpha} > \frac{b}{c} \iff bc - c\alpha\eta > bc - b\alpha \iff b > c\eta \iff b/c > \eta, \quad (\text{A6})$$

which can never hold. Thus, the converse is always true and it holds with equality at $b/c = \eta$.

- (2) The full-cooperative efforts of the countries with reciprocal preferences are higher than the ones with self-interested preferences if

$$\begin{aligned}
\frac{Nb + \alpha(1-2\eta)}{c-2\alpha} > \frac{Nb}{c} = 1 &\iff c + \alpha(1-2\eta) > c-2\alpha \iff \\
1-2\eta > -2 &\iff 3/2 > \eta,
\end{aligned}$$

which always holds. But since $q_{FC}^S = \frac{Nb}{c} = 1$, the upper boundary of the strategy space binds and, thus, $q_{FC}^R = 1$.

■

Proof of Proposition 2.

(1) Let us take the derivative of q_s^R with respect to k :

$$\frac{\partial q_s^R}{\partial k} = \frac{(bc(N-1) + c(1-\eta))(c^2(N-1) - \alpha c(N+k-3) - 2\alpha^2))}{(c^2(N-1) - \alpha c(N+k-3) - 2\alpha^2)^2} + \frac{\alpha c(bck(N-1) + \alpha(bN + \alpha(1-2\eta) + c(k-1 - \eta(N+k-2))))}{(c^2(N-1) - \alpha c(N+k-3) - 2\alpha^2)^2} \quad (\text{A7})$$

We employ Mathematica to algebraically simplify this term and obtain the following:

$$\frac{\partial q_s^R}{\partial k} = \frac{c(c(N-1) + \alpha)[bc(N-1) - \alpha b(N-2) - \alpha(\alpha + c(\eta-1))]}{[c^2(N-1) - \alpha c(N+k-3) - 2\alpha^2]^2} \quad (\text{A8})$$

Next, we show that q_s^R increases in k . All the terms except $\phi \equiv bc(N-1) - \alpha b(N-2) - \alpha(\alpha + c(\eta-1))$ is clearly positive. This term can be reduced to $[b(N-1) + \alpha(1-\eta)] \left[c - \alpha \left(1 - \frac{b-\alpha\eta}{b(N-1) + \alpha(1-\eta)} \right) \right]$.²² The first bracket is clearly positive. The second bracket is also positive by our assumptions $c > 2\alpha$ and $b \geq \alpha\eta$, implying $c > \alpha > \alpha \left(1 - \frac{b-\alpha\eta}{b(N-1) + \alpha(1-\eta)} \right)$. Therefore, we conclude that $\frac{\partial q_s^R}{\partial k} > 0$.

Since $\frac{dq_n}{dq_s} = \frac{\alpha k}{c(N-1) - \alpha(N-k-1)} > 0$ for all $k > 0$, we have

$$\frac{dq_n}{dq_s} = \frac{\partial q_n^R / \partial k}{\partial q_s^R / \partial k} > 0.$$

Then, $\frac{\partial q_s^R}{\partial k} > 0 \implies \frac{\partial q_n^R}{\partial k} > 0$, concluding the proof of point 1.

(2) First, we shall show $q_{NC}^R \leq q_n^R \iff q_n^R \leq q_s^R$. Then we will show $q_n^R \leq q_s^R$ so that $q_{NC}^R \leq q_n^R \leq q_s^R$ holds.

²²Let us show the details by starting from the end:

$$\begin{aligned} & [b(N-1) + \alpha(1-\eta)] \left[c - \alpha \left(1 - \frac{b-\alpha\eta}{b(N-1) + \alpha(1-\eta)} \right) \right] \\ &= bc(N-1) + \alpha c(1-\eta) - \alpha \left(b(N-1) + \alpha(1-\eta) - \frac{(b-\alpha\eta)(b(N-1) + \alpha(1-\eta))}{b(N-1) + \alpha(1-\eta)} \right) \\ &= bc(N-1) + \alpha c(1-\eta) - \alpha b(N-1) - \alpha^2(1-\eta) + \alpha b - \alpha^2\eta \\ &= bc(N-1) - \alpha b(N-2) - \alpha(\alpha - \alpha\eta - c(1-\eta) + \alpha\eta) \\ &= bc(N-1) - \alpha b(N-2) - \alpha(\alpha + c(\eta-1)). \end{aligned}$$

Since $q_{NC}^R = \frac{b-\alpha\eta}{c-\alpha}$, $q_{NC}^R \leq q_n^R$ can be expressed as

$$\begin{aligned}
q_{NC}^R &= \frac{b-\alpha\eta}{c-\alpha} \leq q_n^R \\
&\Leftrightarrow b-\alpha\eta \leq (c-\alpha)q_n^R \\
&\Leftrightarrow b-\alpha\eta \leq \left(c - \alpha \frac{N-k-1}{N-1} - \alpha \frac{k}{N-1}\right) q_n^R \\
&\Leftrightarrow \alpha \frac{k}{N-1} q_n^R \leq \left(c - \alpha \frac{N-k-1}{N-1}\right) q_n^R - (b-\alpha\eta) \\
&\Leftrightarrow q_n^R \leq \frac{\left(c - \alpha \frac{N-k-1}{N-1}\right) q_n^R - (b-\alpha\eta)}{\frac{\alpha k}{N-1}}
\end{aligned}$$

Since $q_n^R = \frac{dq_n}{dq_s} q_s^R + \frac{(N-1)(b-\alpha\eta)}{c(N-1)-\alpha(N-k-1)}$, we can rewrite q_s^R using q_n^R : $q_s^R = \frac{q_n^R - \frac{(N-1)(b-\alpha\eta)}{c(N-1)-\alpha(N-k-1)}}{\frac{dq_n}{dq_s}}$. We know that $\frac{dq_n}{dq_s} = \frac{\alpha k}{c(N-1)-\alpha(N-k-1)}$, so $q_s^R = \frac{q_n^R - \frac{(N-1)(b-\alpha\eta)}{c(N-1)-\alpha(N-k-1)}}{\frac{\alpha k}{c(N-1)-\alpha(N-k-1)}} = \frac{(c-\alpha \frac{N-k-1}{N-1})q_n^R - (b-\alpha\eta)}{\frac{\alpha k}{N-1}}$, which is exactly the right-hand side of the inequality above. Hence, $q_{NC}^R \leq q_n^R \Leftrightarrow q_n^R \leq q_s^R$.

Next, we shall show $q_n^R \leq q_s^R$. Using the equations (13) and (15), we can write down $q_s^R - q_n^R$ (details in the Mathematica file (and its PDF version)) and study its sign.

$$\begin{aligned}
q_s^R - q_n^R &= \frac{(N-1)[c(k-1)+\alpha][bc(N-1)-b(N-2)\alpha-\alpha(\alpha+c(\eta-1))]}{[c(N-1)-\alpha(N-k-1)][c^2(N-1)-c\alpha(N+k-3)-2\alpha^2]} \\
&= \frac{(N-1)[c(k-1)+\alpha]\phi}{[c(N-1)-\alpha(N-k-1)]\Phi}.
\end{aligned} \tag{A9}$$

We showed in part (1) of the proof that $\phi > 0$. Let us also show that $\Phi \equiv [c^2(N-1) - c\alpha(N+k-3) - 2\alpha^2] > 0$. Adding and subtracting $\alpha(c(N-1)+2\alpha)$ to Φ and reorganizing the terms yield $\Phi = (c(N-1)+2\alpha)(c-2\alpha) + \alpha(c(N-k-1)+2\alpha)$. Given the assumption $c > 2\alpha$, all the terms are positive and, thus, $\Phi > 0$.

Thus, $q_s^R - q_n^R = \frac{(N-1)[c(k-1)+\alpha]\phi}{[c(N-1)-\alpha(N-k-1)]\Phi} \geq 0$, which concludes the proof of point 2. ■

Proof of Proposition 3.

Using Mathematica, we find the partial derivative of signatory effort with respect to $\underline{\alpha}$, then evaluated $\underline{\alpha} \approx 0$ and, finally, searched below a condition for this term to be positive:

$$\begin{aligned}
\frac{\partial q_s^R}{\partial \underline{\alpha}}|_{\underline{\alpha}=0} &= -\frac{b[N+k(N+k-3)] + c[k-1-\eta(N+k-2)]}{(b-c)c(N-1)} > 0 \\
&\Leftrightarrow b[N+k(N+k-3)] + c[k-1-\eta(N+k-2)] > 0 \\
&\Leftrightarrow \eta < \frac{b[N+k(N+k-3)] + c[k-1]}{c(N+k-2)} = T_H.
\end{aligned}$$

The minus above is cancelled as we remove the negative denominator since $b < c$ by assumption. Also, if $\eta \geq T_H$, then $\frac{\partial q_s^R}{\partial \underline{\alpha}}|_{\underline{\alpha}=0} \leq 0$.

Analyzing q_n^R in a similar way, we obtain

$$\begin{aligned}\frac{\partial q_n^R}{\partial \alpha}\bigg|_{\alpha=0} &= \frac{b(N + k(k-1) - 1) - c\eta(N-1)}{c(c-b)(N-1)} \\ &\Leftrightarrow \eta < \frac{b(N + k(k-1) - 1)}{c(N-1)} = T_L.\end{aligned}$$

Also, if $\eta \geq T_L$, then $\frac{\partial q_n^R}{\partial \alpha}\big|_{\alpha=0} \leq 0$.

Next, again by employing Mathematica, we find $T_H > T_L \iff$

$$\frac{bk}{k-1} > \frac{bk}{N-1} + \frac{2b-c}{N+k-2}$$

Dividing both sides by c , multiplying by N , and using the assumption that $bN/c = 1$ yields:

$$\frac{k}{k-1} - \frac{k}{N-1} > \frac{2-N}{N+k-2},$$

which always hold given that the LHS is non-negative and the RHS is negative.

Let us also check the conditions so that these three thresholds lie in $[\frac{b}{c}, 1]$:

- First we find that $b/c < T_L < 1$ for $k \leq N-1$ and it exceeds 1 if $k = N$,
(the condition on the lower boundary, $T_L > b/c$, is immediate and omitted here):

$$\begin{aligned}\frac{b(N + k(k-1) - 1)}{c(N-1)} > 1 &\iff bN/c > \frac{N(N-1)}{N+k(k-1)-1} \iff \\ N + k(k-1) - 1 > N^2 - N &\iff 0 > N^2 - 2N - k(k-1) + 1\end{aligned}$$

Notably, the RHS is decreasing in k . For $k = N$, the condition becomes:

$$\begin{aligned}0 &> N^2 - 2N - N(N-1) + 1 \\ 0 &> N^2 - 2N - N^2 + N \\ 0 &> -N\end{aligned}$$

But for $k = N-1$:

$$\begin{aligned}0 &< N^2 - 2N - (N-1)(N-2) + 1 \\ 0 &< N^2 - 2N - N^2 + 3N - 1 \\ 0 &< N - 1\end{aligned}$$

- $T_H > 1 \iff$

$$b(N + k(N + k - 3)) + c(k-1) > c(N + k - 2)$$

Multiplying both sides of this inequality by N/c and using $bN/c = 1$:

$$\begin{aligned}N + k(N + k - 3) + N(k-1) &> N(N + k - 2) \\ k(N + k - 3) &> N(N - 2) \\ k(N + k - 3) - N(N - 2) &> 0\end{aligned}$$

Solving for k yields for any $k > \frac{3-N+\sqrt{(N-1)(5N-9)}}{2}$, $T_H > 1$.

Finally, since $T_L > b/c$ and $T_H > T_L$, we have $T_H > b/c$.

■

Proof of Lemma 1.

First of all, we use the indirect utility functions presented in our supporting Mathematica file (and its PDF version).

- (1) Observe from our Mathematica file that

$$\frac{\partial U_s}{\partial k} = \left(k - 1 + \frac{\alpha}{c}\right) \cdot \underbrace{\frac{\frac{1}{c} \left(n - 1 + \frac{\alpha}{c}\right)^2 \left[b \left(n - 1 - (n - 2) \frac{\alpha}{c} - \frac{\alpha^2}{c} + \alpha(1 - \eta)\right)\right]^2}{\left[\frac{\alpha}{c}k + (n - 1) \left(1 - \frac{\alpha}{c}\right)\right]^2 \left[-\frac{\alpha}{c}k + (n - 1) - (n - 3) \frac{\alpha}{c} - 2 \left(\frac{\alpha}{c}\right)^2\right]^2}}_{\text{positive}}$$

Solving for $\frac{\partial U_s}{\partial k} = 0$ in k yields $k = 1 - \frac{\alpha}{c}$. This k value, denoted by z^{min} , minimizes the signatories' indirect utility.

- (2) We can see that if $k > z^{min} = 1 - \frac{\alpha}{c}$, U_s increases in k and decreases in k otherwise.
 (3) Expressing the non-signatories' indirect utility with the signatories' utility, we have

$$U_{ns} = U_s + \left(k - 1 + \frac{\alpha}{c}\right) \cdot \frac{c \left(c^2(k + 1)(N - 1)^2 - \alpha c(N - 2k - 3)(N - 1) - 2\alpha^2(N - k - 1)\right)}{2(c(N - 1) - \alpha(N - k - 1))^2 (-c^2(N - 1) + \alpha c(N + k - 3) + 2\alpha^2)^2} \cdot \frac{(b(c - cN + \alpha(N - 2)) + \alpha(\alpha + c(\eta - 1)))^2}{2(c(N - 1) - \alpha(N - k - 1))^2 (-c^2(N - 1) + \alpha c(N + k - 3) + 2\alpha^2)^2}$$

Thus, we have $\frac{U_{ns} - U_s}{(k - z^{min})} > 0 \iff$

$$c^2(k + 1)(N - 1)^2 - \alpha c(N - 2k - 3)(N - 1) - 2\alpha^2(N - k - 1) > 0 \quad (\text{A10})$$

We will show that this condition always holds. First taking derivative of this term on the LHS with respect to c and simplifying with Mathematica yields:

$$(N - 1) (2c(k + 1)(N - 1) - \alpha(N - 2k - 3)) \quad (\text{A11})$$

Thus, c increases and α decreases how the term A10 increase in c . Assuming that c takes the smallest possible value, $c = 2\alpha$, since we assume $c > 2\alpha$. (A11) becomes:

$$\alpha(N - 1)(3N - 1 + k(4N - 2)) > 0,$$

which is clearly positive and, thus, the term (A10) increases in c . So, the smallest it can be is at $c = 2\alpha$. Substituting it into the term (A10) yields:

$$2(k + (N - 1)N + 2k(n - 1)N)\alpha^2 > 0,$$

which is always positive. Therefore, the sign of $(U_{ns} - U_s)$ depends only on k 's location compared to z^{min} .

- (4) By the equation (A10), we can conclude $U_{ns}(z^{min}) = U_s(z^{min})$.
- (5) By point (4), we showed that $U_{ns}(z^{min}) = U_s(z^{min})$. By point (3), $U_{ns}(k) > U_s(k)$ for all $k > z^{min}$. Finally, by point (2), since $U_s(k)$ increases in k if $k > z^{min}$, we have $U_{ns}(k)$ increases for $k > z^{min}$.

■

Proof of Lemma 2. The detailed calculations are in the supporting Mathematica file (and its PDF version).

- (1) We can express the impact of a small reciprocal concern (α) on the U_s as

$$\begin{aligned} \left. \frac{\partial U_s(k)}{\partial \alpha} \right|_{\alpha=0} &= \left. \frac{\partial U_s}{\partial q_s^R} \frac{\partial q_s^R}{\partial \alpha} \right|_{\alpha=0} + \left. \frac{\partial U_s}{\partial q_n^R} \frac{\partial q_n^R}{\partial \alpha} \right|_{\alpha=0} \\ &= \frac{b(N-k)}{(c-b)c(N-1)} (b(N+k(k-1)-1) - c\eta(N-1)) \\ \left. \frac{\partial U_s(k)}{\partial \alpha} \right|_{\alpha=0} &\begin{matrix} \geq \\ \leq \end{matrix} 0 \iff \eta \begin{matrix} \leq \\ \geq \end{matrix} T_L = \frac{b(N+k(k-1)-1)}{c(N-1)} \end{aligned} \quad (\text{A12})$$

- (2) We can express the impact of a small α on the U_{ns} as

$$\begin{aligned} \left. \frac{\partial U_{ns}(k)}{\partial \alpha} \right|_{\alpha=0} &= \left. \frac{\partial U_{ns}}{\partial q_s^R} \frac{\partial q_s^R}{\partial \alpha} \right|_{\alpha=0} + \left. \frac{\partial U_{ns}}{\partial q_n^R} \frac{\partial q_n^R}{\partial \alpha} \right|_{\alpha=0} \\ &= b \frac{b((N-1)^2 + k^2(2N-3) - k(N-2)) + c(k(1-\eta)(k-1) - \eta(N-1)^2)}{(c-b)c(N-1)} \end{aligned} \quad (\text{A13})$$

$$\begin{aligned} \left. \frac{\partial U_{ns}(k)}{\partial \alpha} \right|_{\alpha=0} &\begin{matrix} \geq \\ \leq \end{matrix} 0 \iff \\ \eta &\begin{matrix} \leq \\ \geq \end{matrix} T_{U_{ns}} = \frac{ck(k-1) + b((N-1)^2 + k^2(2N-3) - k(N-2))}{c((N-1)^2 + k(k-1))} \end{aligned}$$

- (3) For $U_{ns}(k-1)$ instead of $U_{ns}(k)$, the impact of α could be written as

$$\begin{aligned} \frac{\partial U_{ns}(k-1)}{\partial \alpha} &= \frac{2b(N-k)}{c^2(N-1)} (b(N+k^2-3k+1) - c\eta(N-1)) \\ \left. \frac{\partial U_{ns}(k-1)}{\partial \alpha} \right|_{\alpha=0} &\begin{matrix} \geq \\ \leq \end{matrix} 0 \iff \\ \eta &\begin{matrix} \leq \\ \geq \end{matrix} T_{U_{ns}(k-1)} = \frac{c(k-2)(k-1) + b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} \end{aligned} \quad (\text{A14})$$

- (4) Comparison of Thresholds:

- by Mathematica, we have $T_{U_{ns}} > T_L$ if

$$\frac{k(2b(N-1) + c(k-1)(N-1) + bk(N(N-4) - k(k-2) + 2))}{c((N-1)^2 + k(k-1))(N-1)} > 0$$

Since the denominator is positive, let us focus on the sign of the numerator. Dividing it first by $k > 0$, then multiplying it by $N/c > 0$ allows us to get rid of b and c , using $bN/c = 1$:

$$(N-1)(2 + N(k-1)) + k(N(N-4) - k(k-2) + 2) > 0$$

Given the first term is positive for $N \geq k > 1$, if $N-4 \geq k-2 \iff N \geq k+2$, then the condition holds. If $N = k+1$, the condition also holds:

$$\begin{aligned} (N-1)(2 + N(N-2)) + (N-1)(N(N-4) - (N-1)(N-3) + 2) &> 0 \\ (N-1)(2 + N^2 - 2N + N^2 - 4N - N^2 + 4N - 3 + 2) &> 0 \\ N^2 - 2N + 1 &= (N-1)^2 > 0 \end{aligned}$$

Finally, if $N = k$, the condition holds as long as $N > 2$:

$$\begin{aligned} (N-1)(2 + N(N-1)) + N(N(N-4) - N(N-2) + 2) &> 0 \\ N^3 - 4N^2 + 5N - 2 &= (N-1)^2(N-2) > 0 \end{aligned}$$

- We have $T_{U_{ns}} > T_{U_{ns}(k-1)}$ if

$$\frac{(N-1)(2c(k-1)(N-1) - b(k^2 - k(3-2N)^2 + (N-1)(3N-7)))}{c(k(k-1) + (N-1)^2)(N(N-2) + k(k-3) + 3)} > 0$$

Since the denominator is positive, let us focus on the numerator being positive:

$$\begin{aligned} 2c(k-1)(N-1) &> b(k^2 - k(3-2N)^2 + (N-1)(3N-7)) \iff \\ 2N(k-1)(N-1) &> \underbrace{\frac{bN}{c}}_{=1} (k^2 - k(3-2N)^2 + (N-1)(3N-7)) \iff \\ 2N(k-1)(N-1) &> k^2 - k(3-2N)^2 + (N-1)(3N-7) \end{aligned}$$

By Mathematica, this term is simplified to:

$$N(8 - 3N) + k(7 + 2N(2N - 5)) > 5 + k^2,$$

which always holds for $k \geq 1$.

- By Mathematica, we have $T_L > T_{U_{ns}(k-1)}$ if

$$\frac{(k-1)(-c(k-2)(N-1) + b(N^2(k-3) + N(4k-10) + k(k(k-3) - 1) + 7))}{c(N-1)(N(N-2) + k(k-3) + 3)} > 0$$

Since the denominator is positive, let us focus on the numerator being positive. Dividing it by $k-1 > 0$, taking the term starting with c to the RHS, then multiplying both sides by N/c allow us to get rid of b and c using $bN/c = 1$:

$$\begin{aligned} b(N^2(k-3) + N(4k-10) + k(k(k-3)-1) + 7) &> c(k-2)(N-1) \\ N^2(k-3) + N(4k-10) + k(k(k-3)-1) + 7 &> N(k-2)(N-1) \end{aligned}$$

Further simplification in Mathematica yields:

$$N^2(5-2k) + N(5k-12) + k(k-3) + 7 > 0,$$

which holds for $k \leq 2$ and the converse, $T_L < T_{U_{ns}(k-1)}$, holds if $k \geq 3$.

- Let us also check the conditions so that these three thresholds lie in $[\frac{b}{c}, 1]$:

– First, since $T_{U_{ns}} > T_L > \frac{b}{c}$, the lower bound is satisfied. $T_{U_{ns}} < 1$ if

$$\begin{aligned} \frac{ck(k-1) + b((N-1)^2 + k^2(2N-3) - k(N-2))}{c((N-1)^2 + k(k-1))} &< 1 \\ \frac{b((N-1)^2 + k^2(2N-3) - k(N-2))}{c((N-1)^2 + k(k-1))} &< 1 - \frac{k(k-1)}{(N-1)^2 + k(k-1)} \\ \frac{b((N-1)^2 + k^2(2N-3) - k(N-2))}{c((N-1)^2 + k(k-1))} &< \frac{(N-1)^2}{(N-1)^2 + k(k-1)} \\ \frac{(N-1)^2 + k^2(2N-3) - k(N-2)}{(N-1)^2 + k(k-1)} &< \frac{N(N-1)^2}{(N-1)^2 + k(k-1)} \\ (N-1)^2 + k^2(2N-3) - k(N-2) &< N(N-1)^2 \\ (N-1)^3 - k^2(2N-3) + k(N-2) &> 0 \end{aligned}$$

This shows that the condition holds for not so high k relative to N , but for a given N it binds for sufficiently high k . For instance, the condition does not hold at $N = 10$ and $k = 7$, thus $T_{U_{ns}} > 1$.

– $T_{U_{ns}(k-1)} > b/c$ if

$$\begin{aligned} \frac{c(k-2)(k-1) + b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} &> \frac{b}{c} \\ \frac{b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} &> \frac{b}{c} - \frac{(k-2)(k-1)}{N(N-2) + k(k-3) + 3} \\ \frac{N^2 + kN(2k-5) + N + k(8-3k) - 4}{N(N-2) + k(k-3) + 3} &> 1 - \frac{N(k-2)(k-1)}{N(N-2) + k(k-3) + 3} \\ N^2 + kN(2k-5) + N + k(8-3k) - 4 &> N(N-2) + k(k-3) + 3 - N(k-2)(k-1) \\ (k-1)(N(3k-5) - 4k + 7) &> 0 \end{aligned}$$

This condition holds for $k > 1$.

And $T_{U_{ns}(k-1)} < 1$ if

$$\begin{aligned}
& \frac{c(k-2)(k-1) + b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} < 1 \\
& \frac{b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} < 1 - \frac{(k-2)(k-1)}{N(N-2) + k(k-3) + 3} \\
& \frac{b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} < \frac{N(N-2) + k(k-3) + 3 - (k-2)(k-1)}{N(N-2) + k(k-3) + 3} \\
& \frac{N^2 + kN(2k-5) + N + k(8-3k) - 4}{N(N-2) + k(k-3) + 3} < \frac{N^2(N-2) + 5N}{N(N-2) + k(k-3) + 3} \\
& N^2 + kN(2k-5) + N + k(8-3k) - 4 < N^2(N-2) + 5N \\
& 0 < N^2(N-3) + N(5k+4) - k^2(2N-3) - 8k
\end{aligned}$$

This condition clearly decreasing in k . And for sufficiently high k , the converse of the condition holds. For instance, the condition does not hold for $N = 10$ and $k = 8$, meaning that the threshold exceeds 1. ■

Proof of Proposition 4. For self-interested preferences, Barrett (1994) shows that if $N = 2$, then $k^* = 2$ and for $N \geq 3$, $k^* = 3$ with the signatories being indifferent between joining or staying out, $U_s(z) = U_{ns}(z-1)$. Given the continuity of indirect utility functions and α sufficiently small, the stable coalition size has to be either 2 or 3. In fact, $\eta \in (T_L, T_{U_{ns}(k-1)})$ is sufficient for the coalition to contract to 2, since at this interval U_s is decreasing and $U_{ns}(k-1)$ increasing and thus $U_s(k) = U_{ns}(k-1)$ at a $k < 3$.

The stable coalition size could also shrink in $\eta \in (\frac{b}{c}, T_L)$ in which all indirect utility functions are increasing, if $U_{ns}(k-1)$ increases faster than $U_s(k)$ around $k = 3$:

$$\begin{aligned}
& \left. \frac{\partial U_s(k)}{\partial \alpha} \right|_{\alpha=0, k=3} < \left. \frac{\partial U_{ns}(k-1)}{\partial \alpha} \right|_{\alpha=0, k=3} \\
& \frac{b(3-N)(b(N+5) - c\eta(N-1))}{(b-c)c(N-1)} < \frac{b(7b-2c - bN(N+4) + c\eta(3+N(N-2)))}{(b-c)c(N-1)} \\
& b(3-N)(b(N+5) - c\eta(N-1)) > b(7b-2c - bN(N+4) + c\eta(3+N(N-2)))
\end{aligned}$$

The direction of the inequality is reversed because the denominator is negative as $b < c$. Simplifying it further using Mathematica (see also its PDF version) yields:

$$\begin{aligned}
& b(c + b(N+4) - cN\eta) > 0 \iff b(N+4) > c(\eta N - 1) \iff N+4 > \eta N^2 - N \\
& \iff \eta < \frac{2N+4}{N^2} \equiv T_2
\end{aligned}$$

Thus, coalition size contracts to 2 if η is smaller than T_2 .

Next, we show that $T_2 > T_{U_{ns}(k-1)} \iff$

$$\begin{aligned} \frac{2N+4}{N^2} &> \frac{c(k-2)(k-1) + b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} \\ \frac{2N+4}{N^2} - \frac{(k-2)(k-1)}{(N(N-2) + k(k-3) + 3)} &> \frac{b(N^2 + kN(2k-5) + N + k(8-3k) - 4)}{c(N(N-2) + k(k-3) + 3)} \\ \frac{2N+4}{N} - \frac{N(k-2)(k-1)}{(N(N-2) + k(k-3) + 3)} &> \frac{N^2 + kN(2k-5) + N + k(8-3k) - 4}{N(N-2) + k(k-3) + 3} \end{aligned}$$

Employing Mathematica to simplify this term yields:

$$2 + \frac{4}{N} > \frac{(N-1)(N + k(3k-8) + 4)}{N(N-2) + k(k-3) + 3}$$

Evaluating this condition around $k = 3$ yields:

$$\frac{(N-4)(N-3)(N+1)}{N(N(N-2) + 3)} > 0, \quad (\text{A15})$$

which always holds for $N \geq 5$.

Next, we show that $T_2 > T_{U_{ns}} \iff$

$$\begin{aligned} \frac{2N+4}{N^2} &> \frac{ck(k-1) + b((N-1)^2 + k^2(2N-3) - k(N-2))}{c((N-1)^2 + k(k-1))} \\ \frac{2N+4}{N^2} - \frac{k(k-1)}{(N-1)^2 + k(k-1)} &> \frac{b((N-1)^2 + k^2(2N-3) - k(N-2))}{c((N-1)^2 + k(k-1))} \\ \frac{2N+4}{N} - \frac{Nk(k-1)}{(N-1)^2 + k(k-1)} &> \frac{(N-1)^2 + k^2(2N-3) - k(N-2)}{(N-1)^2 + k(k-1)} \end{aligned}$$

Employing Mathematica to simplify this term yields:

$$2 + \frac{4}{N} > \frac{(N-1)(N + k(3k-2) - 1)}{(N-1)^2 + k(k-1)}$$

Evaluating this condition around $k = 3$ yields:

$$\frac{N^3 - 19N^2 + 26N + 28}{N(N^2 - 2N + 7)} > 0. \quad (\text{A16})$$

Notice that the denominator is always positive. The numerator is positive and the condition holds for $N \geq 18$, but it is negative and $T_2 < T_{U_{ns}}$ holds for $N \leq 17$.

Next, we study the condition for $T_H > T_2$. Simplifying the condition in Mathematica yields:

$$-\frac{2(N+2)}{N^2} + \frac{c(k-1) + b(N + k(N+k-3))}{c(N+k-2)} > 0 \quad (\text{A17})$$

Multiplying both sides by N and use the assumption $bN/c = 1$ yields:

$$\frac{N(k-1) + (N + k(N+k-3))}{(N+k-2)} > \frac{2(N+2)}{N}$$

Evaluating this condition around $k = 3$ yields:

$$(N-2)N(N+1)(1+2N) > 0, \quad (\text{A18})$$

which always hold for $N \geq 3$.

Finally, we study the condition for $T_H > T_{U_{ns}}$. Simplifying the condition in Mathematica yields:

$$\frac{(k+1-N)(c(-kN+N+k-1)+b(2-2N+k(N(N-4)+k(k-2)-2)))}{c(k(k-1)+(N-1)^2)(N+k-2)} > 0$$

Notice that the denominator is positive and there are two parentheses multiplied in the numerator. Let us focus on the sign of the second one, which is negative if:

$$\begin{aligned} c(-kN+N+k-1)+b(2-2N+k(-N(N-4)+k(k-2)-2)) &< 0 \iff \\ 2-2N+k(-N(N-4)+k(k-2)-2) &< N(kN-N-k+1) \end{aligned}$$

Since we are interested in the order around $k = 3$, we substitute $k = 3$ into the inequality and simplify:

$$\begin{aligned} 2-2N+3(-N^2+4N+3-2) &< N(3N-N-3+1) \\ 2-2N-3N^2+12N+3 &< 2N^2-2N \\ -5N^2+12N+5 &< 0 \end{aligned}$$

Solving this inequality for N , we find that it holds if $N > 2.76205$. Since the first parenthesis in the numerator is also negative for $N > k + 1$. For $N \geq 5$, we have $T_H > T_{U_{ns}}$ at $k \approx 3$. ■

Proof of Proposition 5. First notice that self-interested countries have still dominant strategy to exert $q_{NC}^{HS} = b/c$ in the non-cooperative game with heterogeneous countries. Thus, a reciprocal country i maximizes the following utility function while $A - 1$ other reciprocal country j 's exert q_j :

$$U_{Hi} = b((N-A)\frac{b}{c} + (A-1)q_j + q_i) - \frac{c}{2}q_i^2 + \alpha \left(\frac{(N-A)\frac{b}{c} + (A-1)q_j}{N-1} - \eta \right) (q_i + 1 - \eta)$$

Finding the F.O.C., equating $q_{NC}^{HR} = q_i = q_j$, and solving for q_{NC}^{HR} yields the Nash efforts:

$$q_{NC}^{HR} = \frac{bc(N-1) + b(N-A)\alpha - c(N-1)\alpha\eta}{c(c(N-1) + \alpha - A\alpha)} \quad (\text{A19})$$

Taking derivative of the Nash efforts of the reciprocal countries with respect to α yields the impact of reciprocal concerns:

$$\frac{\partial q_{NC}^{HR}}{\partial \alpha} = \frac{(N-1)^2(b - c\eta)}{(c(N-1) - \alpha(A-1))^2} \quad (\text{A20})$$

Thus, $\frac{\partial q_{NC}^{HR}}{\partial \alpha} \leq 0 \iff \eta \geq \frac{b}{c} = q_{NC}^{HS}$.

Substituting $b = c/N$ and solving the condition $q_{NC}^{HR} \leq 0$ for α yields

$$q_{NC}^{HR} \leq 0 \iff \alpha \geq \frac{c(N-1)}{A + N(\eta(N-1) - 1)} \iff \underline{\alpha} \geq \frac{c(N-1)^2}{N(A + N(\eta(N-1) - 1))}$$

If A and η are large enough, the RHS of the inequality can be less than 1. Thus, it is possible for reciprocal countries to exert no effort. ■